

3.6 Newton's Laws of Motion

Mathematics B (MEI) — OCR B — A-Level (H640) — Mechanics — spec refs n1–n7

What this topic covers. Newton's three laws, forming the equation of motion for a particle, and applying it to connected particles such as pulley systems and trains.

1. The three laws

First law. An object continues in a state of rest or uniform motion in a straight line unless it is acted on by a resultant force.

Second law. A resultant force $\mathbf{F}$ acting on an object of fixed mass m gives the object an acceleration $\mathbf{a}$ given by

$$\mathbf{F} = m\mathbf{a}$$

Third law. When one object exerts a force on another, there is always a reaction equal in magnitude and opposite in direction to the acting force.

The $\mathbf{F}$ in $\mathbf{F} = m\mathbf{a}$ is the **resultant** of all forces, not any single one. Substituting just the applied force, and ignoring friction or weight, is the most common structural error in this topic.

2. The equation of motion

Writing the equation of motion means applying the second law to a particular body in a particular direction (spec refs n2, n3). The method:

1. Draw a force diagram for the body.
2. Choose a positive direction — usually the direction of acceleration.
3. Resolve in that direction and set the total equal to ma .

Worked example. A resultant force of 20 N acts on a 4 kg mass.

$$a = \frac{F}{m} = \frac{20}{4} = 5 \text{ m s}^{-2}$$

Worked example. A car of mass 1000 kg has a driving force of 3000 N and resistance 1000 N.

$$3000 - 1000 = 1000a \implies a = 2 \text{ m s}^{-2}$$

Worked example. A 5 kg body falls against air resistance of 19 N, with $g = 9.8$.

$$5(9.8) - 19 = 5a \implies 49 - 19 = 5a \implies a = 6 \text{ m s}^{-2}$$

Equilibrium is simply the special case $a = 0$, which recovers the condition that the resultant force is zero.

In two dimensions

The second law is a vector equation, so it may be applied component by component (n7).

Worked example. A resultant force $6\mathbf{i} + 8\mathbf{j}$ N acts on a 2 kg particle.

$$\mathbf{a} = 3\mathbf{i} + 4\mathbf{j} \text{ m s}^{-2}, \quad |\mathbf{a}| = 5 \text{ m s}^{-2}$$

Problems involving **variable mass** are excluded from this specification, so m may always be treated as constant.

3. Lifts and apparent weight

For a person of mass m in a lift, the forces are the normal reaction R upwards and the weight mg downwards.

Lift motion	Equation	Reaction
Accelerating upwards at a	$R - mg = ma$	Greater than the weight
Accelerating downwards at a	$mg - R = ma$	Less than the weight
Constant velocity	$R = mg$	Equal to the weight

Worked example. A 70 kg passenger in a lift accelerating upwards at 2 m s^{-2} , with $g = 9.8$.

$$R = m(g + a) = 70(11.8) = 826 \text{ N}$$

A lift *descending* but *decelerating* has its acceleration directed **upwards**, so the reaction exceeds the weight. Read the direction of the acceleration, not the direction of travel.

4. Newton's third law properly understood

A third-law pair acts on **two different bodies**. A book resting on a table: the normal reaction from the table on the book and the book's weight both act on the *book*, so they are **not** a third-law pair — they merely happen to be equal here because the book is in equilibrium. The true partner of the book's weight is the gravitational pull of the book on the Earth.

The test is simple: if two forces act on the same object, they cannot be a third-law pair.

5. Connected particles

A system may be modelled as a set of connected particles — pulley systems, trains, tow ropes (n_4 , n_5).

Two standard assumptions do the work:

- **Light, inextensible string:** the tension is the same throughout, and both particles have accelerations of equal magnitude.
- **Smooth pulley:** the tension is unchanged in passing over it, so it is equal on both sides.

Two strategies

Approach	Use it to find	Why
Whole system as one body	The acceleration	Internal tensions cancel, leaving only external forces
Each particle separately	The tension or coupling force	Internal forces appear explicitly

Worked example. Masses of 5 kg and 3 kg hang from a light inextensible string over a smooth pulley, with $g = 9.8$.

For the 5 kg mass, taking downwards as positive: $5g - T = 5a$.

For the 3 kg mass, taking upwards as positive: $T - 3g = 3a$.

Adding eliminates T :

$$2g = 8a \implies a = \frac{2(9.8)}{8} = 2.45 \text{ m s}^{-2}$$

Substituting back: $T = 3(9.8 + 2.45) = 36.75 \text{ N}$.

Check the tension against the *other* equation: $T = 5(9.8 - 2.45) = 36.75 \text{ N}$. Agreement confirms the arithmetic. Note also that the tension lies **between** the two weights, 29.4 N and 49 N — if yours does not, something is wrong.

Worked example. A 4 kg block on a smooth table is connected over a smooth pulley to a 2 kg hanging mass.

Whole system: the only external driving force is the hanging weight, $2g$, acting on a total mass of 6 kg:

$$2(9.8) = 6a \implies a = 3.27 \text{ m s}^{-2} \text{ (3 s.f.)}$$

For the block alone, $T = 4a = 13.1 \text{ N}$.

Worked example. An engine of mass 20 000 kg pulls a carriage of mass 10 000 kg with a total driving force of 45 000 N and no resistance. Find the coupling force.

Whole system: $45\,000 = 30\,000a$, so $a = 1.5 \text{ m s}^{-2}$.

Carriage alone: the coupling force is the only force on it, so $T = 10\,000 \times 1.5 = 15\,000 \text{ N}$.

Take moments — figuratively — over which body to isolate. To find an internal force, always write the equation for the body on **one side** of it, choosing whichever has fewer forces acting.

6. Terminal velocity

A falling body meets resistance that grows with speed. When the resistance equals the weight, the resultant force is zero, the acceleration becomes zero, and the speed stops increasing — the **terminal velocity**. It is approached asymptotically rather than reached at a definite instant.

7. Common pitfalls

- Using a single force rather than the resultant in $F = ma$.
- Identifying a book's weight and the table's reaction as a third-law pair.
- Using different accelerations for particles joined by an inextensible string.
- Assuming different tensions either side of a smooth pulley.
- Confusing mass and weight in the equation of motion.
- Choosing inconsistent positive directions for the two particles.
- Trying to find an internal tension from the whole-system equation, where it has cancelled.

- Reading the direction of travel instead of the direction of acceleration in a lift problem.